



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2005
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #1

Mathematics Extension 2

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each question is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 85

- Attempt questions 1 – 3
- All questions are not of equal value.

Examiner: *C. Kourtesis*

Question 1. (Start a new answer sheet.) (31 marks)

- | | Marks |
|--|--------------|
| (a) Given that $w = \sqrt{3} + i$, express the following in the form $a + ib$ where a and b are real. | 4 |
| (i) $-iw$ | |
| (ii) w^2 | |
| (iii) w^{-1} | |
| (b) If $z = 1 - i$ find: | 4 |
| (i) $ z $ and $\arg z$ | |
| (ii) z^8 in exact form | |
| (c) Consider the equation | 3 |
| $z^2 + kz + (4 - i) = 0$ | |
| Find the complex number k given that $2i$ is a root of the equation. | |
| (d) If $z = x + iy$ prove that | 3 |
| $z + \frac{ z ^2}{z} = 2 \operatorname{Re}(z)$ | |
| (e) Sketch the locus of z satisfying | 4 |
| (i) $ z + 2i = 2$ | |
| (ii) $\operatorname{Re}(z^2) = 0$ | |
| (f) (i) Plot on the Argand diagram all complex numbers that are roots of $z^5 = 1$. | 4 |
| (ii) Express $z^5 - 1$ as a product of real linear and quadratic factors. | |

- (g) (i) By solving the equation $z^3 + 1 = 0$ find the three cube roots of -1 .
- (ii) Let ω be a cube root of -1 , where ω is not real.
Show that $\omega^2 + 1 = \omega$
- (iii) Hence simplify $(1 - \omega)^{12}$.
- (iv) Find a quadratic equation with real coefficients whose roots are ω^2 and $-\omega$.

Question 2. (Start a new answer sheet.) (31 marks)

Marks

- (a) Given that $\text{cis } \theta = \cos \theta + i \sin \theta$ find in exact form

3

$$\text{cis } \frac{\pi}{12} \text{cis } \frac{\pi}{6}$$

- (b) The equation $x^3 + Ax + B = 0$ (A, B real) has three real roots α , β and γ .

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(i) Evaluate $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ and $\alpha^2 + \beta^2 + \gamma^2$ in terms of A and B .

(ii) Prove that $A < 0$.

(iii) Find the cubic polynomial whose roots are α^2 , β^2 and γ^2 .

- (c) It is given that $z = 1 + i$ is a zero of $P(z) = z^3 + pz^2 + qz + 6$ where p and q are real numbers.

4

(i) Explain why $\bar{z}$ is also a zero of $P(z)$. (State the theorem.)

(ii) Find the values of p and q .

- (d) Find the number of ways in which six women and six men can be arranged in three sets of four for tennis if:

5

(i) there are no restrictions.

(ii) each man has a woman as a partner.

- (e) In the Argand diagram the points O, A and B are the vertices of a triangle with $\angle AOB = 90^\circ$ and $\frac{OB}{OA} = 2$.

6

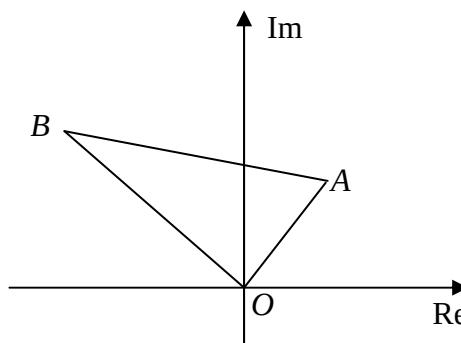
The vertices A and B correspond to the complex numbers z_1 and z_2 respectively.

Show that:

(i) $2z_1 + iz_2 = 0$

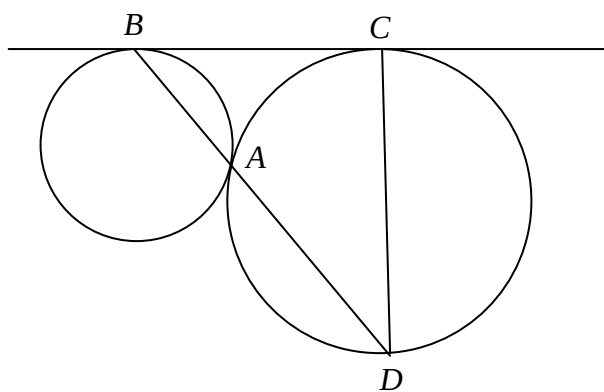
(ii) the equation of the circle with AB as diameter and passing through O is given by

$$\left| z - z_1 \left(\frac{1}{2} + i \right) \right| = \frac{\sqrt{5}}{2} |z_1|.$$



(f)

4



The two circles touch at A and a common external tangent touches them at B and C . BA produced meets the larger circle at D .

Prove that CD is a diameter.

Question 3. (Start a new answer sheet.) (23 marks)

- | | |
|--|--------------|
| | Marks |
| (a) In how many ways can three different trophies be awarded to five golfers if a golfer may receive at most two trophies? | 3 |

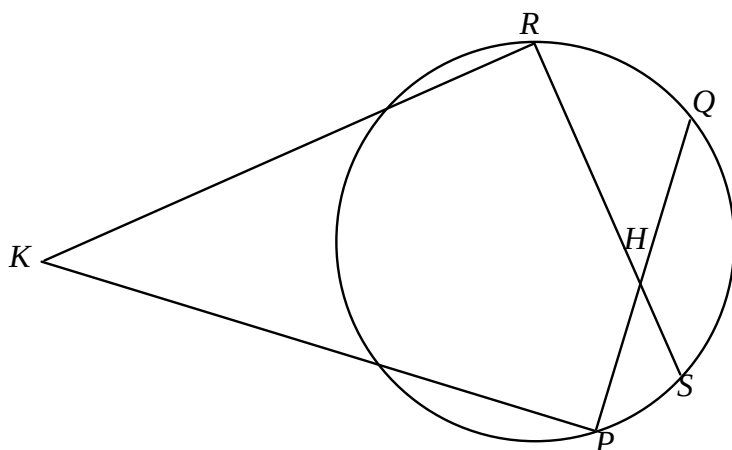
- | | |
|---|---|
| (b) Sketch the region in the Argand diagram consisting of all points z satisfying | 4 |
|---|---|

$$|\arg z| < \frac{\pi}{4} \text{ and } z + \bar{z} < 4 \text{ and } |z| > 2.$$

- | | |
|--|---|
| (c) (i) Prove that $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$, where n is a positive integer. | 6 |
|--|---|

- (ii) Hence or otherwise show that $(1 + z)^4 + (1 - z)^4 = 0$ has roots $\pm i \tan \frac{\pi}{8}$ and $\pm i \tan \frac{3\pi}{8}$

(d)



In the diagram above PQ and RS are two chords intersecting at H , and $\angle KPQ = \angle KRS = 90^\circ$.

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(i) Copy the diagram onto your answer sheet, indicating the above information.

(ii) Prove that $(\alpha) \quad \angle PKH = \angle PQS$.

$(\beta) \quad KH$ produced is perpendicular to QS .

(e) If α is a real root of the equation $x^3 + ux + v = 0$ prove that the other two roots are real if $4u + 3\alpha^2 \leq 0$.

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End of paper.



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Mathematics Extension 2

Sample Solutions

Question	Marker
1	PSP
2	DH
3	PRB

Question 1

(a) $w = \sqrt{3} + i$

(i) $-iw = -i(\sqrt{3} + i) = 1 - i\sqrt{3}$

(ii) $w^2 = (\sqrt{3} + i)^2 = 2 + i2\sqrt{3}$

(iii) $w^{-1} = \frac{\bar{w}}{|w|^2} = \frac{\sqrt{3} - i}{4} = \frac{\sqrt{3}}{4} - i\left(\frac{1}{4}\right)$

(b) $z = 1 - i$

(i) $|z| = \sqrt{2}, \arg(z) = -\frac{\pi}{4}$

(ii) $z^8 = \left(\sqrt{2}\text{cis}\left(-\frac{\pi}{4}\right)\right)^8 = 16\text{cis}\left(-\frac{8\pi}{4}\right) = 16\text{cis}(-2\pi) = 16$

(c) $p(z) = z^2 + kz + (4 - i)$

$$p(2i) = 0 \Rightarrow (2i)^2 + k(2i) + 4 - i = 0$$

$$\therefore -4 + 2ki + 4 - i = 0$$

$$\therefore 2ki = i$$

$$\therefore k = \frac{1}{2}$$

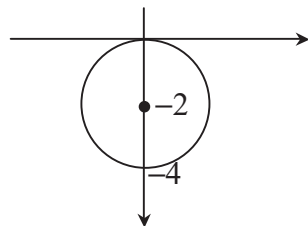
(d) $z = x + iy$

$$\therefore \frac{1}{z} = \frac{\bar{z}}{|z|^2}$$

$$\therefore z + \frac{|z|^2}{z} = z + \bar{z} = 2\text{Re } z$$

(e) (i) $x^2 + (y + 2)^2 = 4$

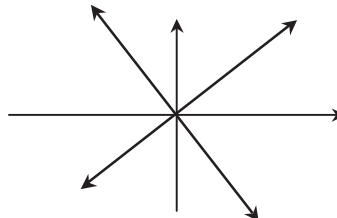
A circle with centre $(0, -2)$ ie $-2i$ and radius 2



(ii) $\text{Re}(z^2) = x^2 - y^2 = 0$

$$\therefore x^2 = y^2$$

$$\therefore y = \pm x$$



$$\begin{aligned}
 (f) \quad (i) \quad z^5 &= 1 \times \text{cis}(0) \\
 &= \text{cis}(0 + 2k\pi), k \in \mathbb{Z} \\
 &= \text{cis}(2k\pi) \\
 z &= [\text{cis}(2k\pi)]^{1/5} \\
 &= \text{cis}\left(\frac{2k\pi}{5}\right) \quad (\text{deMoivre's Theorem})
 \end{aligned}$$

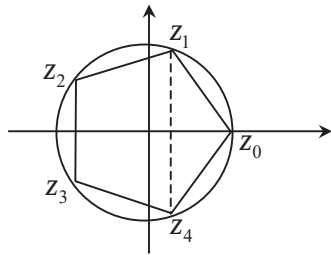
$$k = 0 : \quad z_0 = \text{cis}(0) = 1$$

$$k = 1 : \quad z_1 = \text{cis}\left(\frac{2\pi}{5}\right)$$

$$k = 2 : \quad z_2 = \text{cis}\left(\frac{4\pi}{5}\right) \quad \boxed{|z_k| = 1}$$

$$k = -1 : \quad z_3 = \text{cis}\left(-\frac{2\pi}{5}\right)$$

$$k = -2 : \quad z_4 = \text{cis}\left(-\frac{4\pi}{5}\right)$$



The 5 roots must form a regular pentagon inscribed in a unit circle.

As well:

z_1 and z_4 are conjugates

z_2 and z_3 are conjugates

$$\begin{aligned}
 (ii) \quad (z - \alpha)(z - \bar{\alpha}) &= z^2 - 2\text{Re}(\alpha)z + |\alpha|^2 \\
 z^5 - 1 &= (z - z_0)(z - z_1)(z - z_2)(z - z_3)(z - z_4) \\
 &= (z - 1)(z - z_1)(z - \bar{z}_1)(z - z_2)(z - \bar{z}_2) \\
 &= (z - 1)\left(z^2 - (2\text{Re } z_1)z + |z_1|^2\right)\left(z^2 - (2\text{Re } z_2)z + |z_2|^2\right)z \\
 &= (z - 1)\left(z^2 - 2z \cos \frac{2\pi}{5} + 1\right)\left(z^2 - 2z \cos \frac{4\pi}{5} + 1\right)
 \end{aligned}$$

$$\begin{aligned}
 \text{(g)} \quad \text{(i)} \quad z^3 &= -1 \\
 &= 1 \times \text{cis}(\pi) \\
 &= \text{cis}(\pi + 2k\pi), k \in \mathbb{Z} \\
 &= \text{cis}(2k+1)\pi \\
 z &= [\text{cis}(2k+1)\pi]^{1/3} \\
 &= \text{cis}(2k+1)\frac{\pi}{3} \quad (\text{deMoivre's Theorem})
 \end{aligned}$$

$$k=0: \quad z = \text{cis} \frac{\pi}{3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$k=1: \quad z = \text{cis} \frac{3\pi}{3} = -1$$

$$k=-1: \quad \text{cis}\left(-\frac{\pi}{3}\right) = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$\begin{aligned}
 \text{(ii)} \quad z^3 + 1 &= (z+1)(z^2 - z + 1) \\
 \omega^3 &= -1, \omega \neq -1 \\
 \therefore \omega^3 + 1 &= (\omega+1)(\omega^2 - \omega + 1) = 0 \\
 \therefore \omega^2 - \omega + 1 &= 0 \quad (\because \omega \neq -1) \\
 \therefore \omega^2 + 1 &= \omega
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad (1-\omega)^{12} &= (-\omega^2)^{12} \quad (\text{from (ii)}) \\
 &= (\omega^3)^8 \\
 &= (-1)^8 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad (z - \omega^2)(z + \omega) &= 0 \\
 z^2 + (\omega - \omega^2)z - \omega^3 &= 0 \\
 \therefore z^2 + (1)z - (-1) &= 0 \quad (\text{from (ii)}) \\
 \therefore z^2 + z + 1 &= 0
 \end{aligned}$$

OR more simply since $z^3 + 1 = (z+1)(z^2 - z + 1)$

and the three roots of -1 are so that $z^2 - z + 1 = 0$ must have roots $\omega, -\omega^2$.

So let $y = -z$ and $y^2 + y + 1 = 0$ MUST have roots $-\omega, \omega^2$.

Question 2

(a) Method 1:

3

$$\begin{aligned}\operatorname{cis} \frac{\pi}{12} \operatorname{cis} \frac{\pi}{6} &= \operatorname{cis} \left(\frac{\pi}{12} + \frac{\pi}{6} \right), \text{ by de Moivre's theorem} \\ &= \operatorname{cis} \frac{\pi}{4}, \\ &= \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}.\end{aligned}$$

Method 2:

$$\begin{aligned}\operatorname{cis} \frac{\pi}{12} \operatorname{cis} \frac{\pi}{6} &= \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right) \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right), \\ &= \cos \frac{\pi}{12} \cos \frac{\pi}{6} - \sin \frac{\pi}{12} \sin \frac{\pi}{6} + i \left(\sin \frac{\pi}{12} \cos \frac{\pi}{6} + \cos \frac{\pi}{12} \sin \frac{\pi}{6} \right), \\ &= \cos \frac{\pi}{4} + i \sin \frac{\pi}{4}, \\ &= \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}.\end{aligned}$$

(b) i.

3

$$\begin{aligned}\alpha + \beta + \gamma &= 0, \\ \alpha\beta + \alpha\gamma + \beta\gamma &= A, \\ \alpha\beta\gamma &= -B.\end{aligned}$$

Now, $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma},$

$$= -\frac{A}{B}.$$

$$\begin{aligned}\text{Also, } (\alpha + \beta + \gamma)^2 &= \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma). \\ \therefore \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma), \\ &= 0 - 2A, \\ &= -2A.\end{aligned}$$

ii. Method 1:

2

$$\begin{aligned}A &= -\frac{1}{2}(\alpha^2 + \beta^2 + \gamma^2). \\ \text{But } \alpha^2 + \beta^2 + \gamma^2 &> 0 \text{ if } \alpha \neq \beta \neq \gamma. \\ \therefore A &< 0.\end{aligned}$$

Method 2:

$$P'(x) = 3x^2 + A.$$

If $A > 0$ then $P(x)$ is monotonic increasing so there can be only one real root. But there are 3 real roots so $A < 0$.

iii. Put $X = x^2.$

4

$$\begin{aligned}\therefore x &= \sqrt{X}. \\ X\sqrt{X} + A\sqrt{X} + B &= 0, \\ \sqrt{X}(X + A) &= -B, \\ X(X^2 + 2XA + A^2) &= B^2. \\ \text{New equation is } x^3 + 2Ax^2 + A^2x - B^2 &= 0.\end{aligned}$$

- (c) i. If $a + ib$ is a complex zero of the polynomial $P(x)$ of degree $n \geq 1$ with real coefficients, then $a - ib$ is also a zero of $P(x)$. 1

- ii. Let the roots be $\alpha, 1 + i, 1 - i$, then 3

$$\begin{aligned} z^3 + pz^2 + qz + 6 &= (z - \alpha)(z - 1 - i)(z - 1 + i), \\ &= (z - \alpha)(z^2 - 2z + 2), \\ &= z^3 - (\alpha + 2)z^2 + (2\alpha + 2)z - 2\alpha. \end{aligned}$$

Equating coefficients gives $\alpha = -3$.

$$p = -(-3 + 2),$$

$$= 1.$$

$$q = -6 + 2,$$

$$= -4.$$

- (d) i. There are ${}^{12}C_4$ ways of getting the first group and 8C_4 ways of getting the second group leaving the third group. As the group order does not matter, we have $\frac{{}^{12}C_4 \times {}^8C_4}{3!} = 5775$. 2

- ii. There are ${}^6C_2 \times {}^6C_2$ ways of getting the first and ${}^4C_2 \times {}^4C_2$ ways of getting the second group, leaving the third group. As before, the group order does not matter, so we have $\frac{({}^6C_2 \times {}^4C_2)^2}{3!} = 1350$. 3

Note that we are not asked to arrange the people within the groups, only to form the groups.

- (e) i. Method 1: 2

$$z_2 = 2iz_1 \text{ (Twice the length and rotated anti-clockwise by } 90^\circ\text{),}$$

$$iz_2 = -2z_1,$$

$$\therefore 2z_1 + 2iz_2 = 0.$$

Method 2:

$$\text{Let } z_1 = a + ib,$$

$$z_2 = 2i(a + ib),$$

$$= 2ai - 2b.$$

$$\therefore 2z_1 = 2a + 2bi,$$

$$iz_2 = -2a - 2bi.$$

$$\text{So } 2z_1 + iz_2 = 0.$$

- ii. Method 1: 4

$$\text{Centre} = \frac{z_1 + z_2}{2},$$

$$= \frac{z_1}{2} - \frac{2z_1}{2i} \times \frac{i}{i},$$

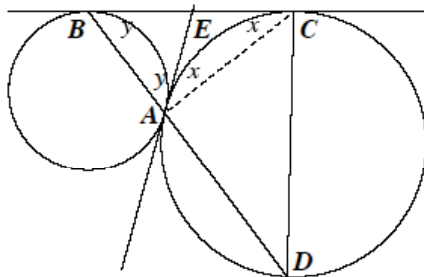
$$= z_1\left(\frac{1}{2} + i\right).$$

$$\begin{aligned}\text{Radius} &= \frac{1}{2}|z_1 - z_2|, \\ &= \frac{1}{2}|z_1 - 2z_1i|, \\ &= \frac{1}{2}|z_1| |1 - 2i|, \\ &= \frac{1}{2}|z_1|\sqrt{1^2 + 2^2}, \\ &= \frac{\sqrt{5}}{2}|z_1|.\end{aligned}$$

Method 2:

$$\begin{aligned}\text{Centre} &= \frac{a-2b}{2} + \frac{i}{2}(b+2a), \\ &= \frac{a+ib}{2} + \frac{2ai-2b}{2}, \\ &= \frac{z_1}{2} + \frac{z_2}{2}, \\ &= \frac{z_1}{2} - \frac{2z_1}{2i} \times \frac{i}{i}, \\ &= z_1\left(\frac{1}{2} + i\right). \\ \text{Radius}^2 &= \left(\frac{a-2b}{2}\right)^2 + \left(\frac{b+2a}{2}\right)^2, \\ &= \frac{a^2 - 4ab + 4b^2 + b^2 + 4ab + 4a^2}{4}, \\ &= \frac{5a^2 + 5b^2}{4}. \\ \text{Radius} &= \frac{\sqrt{5}}{2} \sqrt{a^2 + b^2}, \\ &= \frac{\sqrt{5}}{2} |z_1|\end{aligned}$$

(f) Method 1:



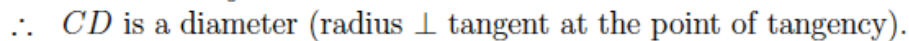
Construct the common tangent at A cutting BC at E .

Join AC .

Let $A\hat{C}E = x$, $E\hat{B}A = y$.

$EC = EA = EB$ (equal tangents from external point),

Method 2:



Question 3

(a) **Method 1:**

Case 1: 3 different golfers receive prizes

$\binom{5}{3}$ picks the golfers and then the prizes can be awarded in $3!$ ways

ie $\binom{5}{3} \times 3! = 60$ ways.

Case 2: 1 golfer receives two prizes

Pick the golfer to receive the prize in $\binom{5}{1}$ ways and his prizes in $\binom{3}{2}$ ways.

Then the remaining prize can go to one of the 4 others

ie $\binom{5}{1} \times \binom{3}{2} \times \binom{4}{1} = 60$ ways

Total = $60 + 60 = 120$

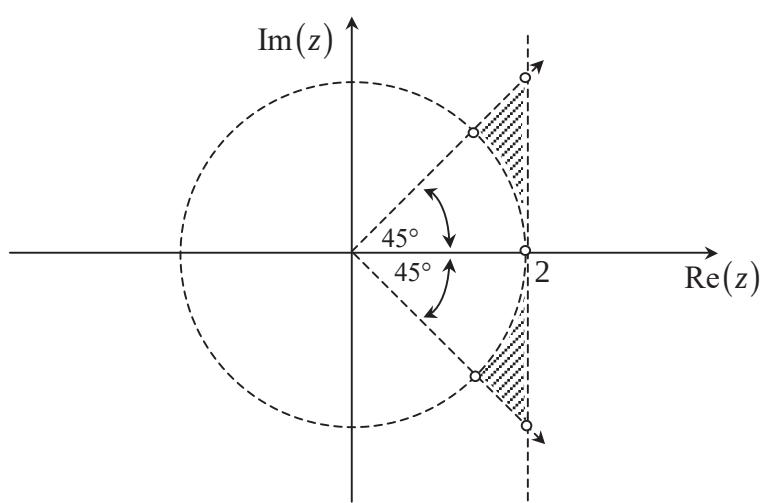
Method 2:

There are $5^3 = 125$ ways of dividing up the prizes with no restrictions.

There are 5 ways in which a golfer can get all the prizes.

So there are $125 - 5 = 120$ ways in dividing up the prizes so that a golfer gets no more than 2 prizes.

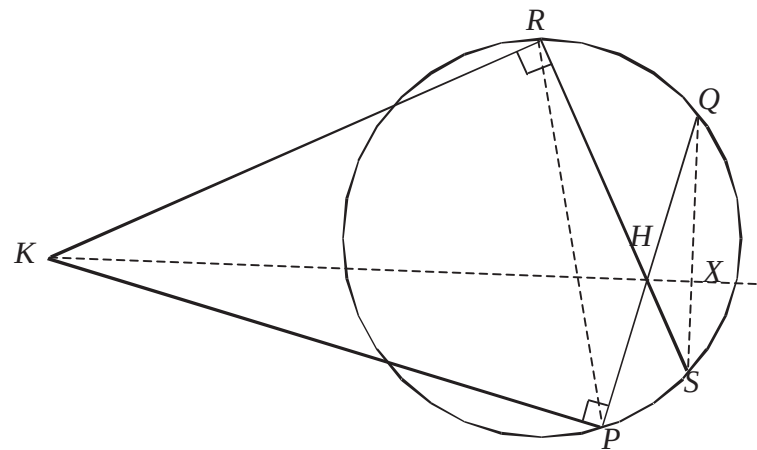
(b) $|\arg z| < \frac{\pi}{4} \Rightarrow -\frac{\pi}{4} < \arg z < \frac{\pi}{4}$
 $z + \bar{z} < 4 \Rightarrow x < 2$



$$\begin{aligned}
(c) \quad (i) \quad \text{LHS} &= (1 + i \tan \theta)^n + (1 - i \tan \theta)^n \\
&= \left(1 + i \frac{\sin \theta}{\cos \theta}\right)^n + \left(1 - i \frac{\sin \theta}{\cos \theta}\right)^n \\
&= \left(\frac{\cos \theta + i \sin \theta}{\cos \theta}\right)^n + \left(\frac{\cos \theta - i \sin \theta}{\cos \theta}\right)^n \\
&= \frac{[\text{cis } \theta]^n + [\text{cis } (-\theta)]^n}{\cos^n \theta} \\
&= \frac{\text{cis } n\theta + \text{cis } (-n\theta)}{\cos^n \theta} \quad (\text{deMoivre's Theorem}) \\
&= \frac{2 \cos n\theta}{\cos^n \theta} \quad (z + \bar{z} = 2 \text{Re } z) \\
&= \text{RHS}
\end{aligned}$$

$$\begin{aligned}
(ii) \quad (1+z)^4 + (1-z)^4 &= \frac{2 \cos 4\theta}{\cos^4 \theta} \text{ where } z = i \tan \theta \text{ from (i)} \\
(1+z)^4 + (1-z)^4 &= 0 \Leftrightarrow \frac{2 \cos 4\theta}{\cos^4 \theta} = 0 \\
\therefore \cos 4\theta &= 0 \\
\therefore 4\theta &= \pm \frac{\pi}{2}, \pm \frac{3\pi}{2} \\
\therefore \theta &= \pm \frac{\pi}{8}, \pm \frac{3\pi}{8} \\
\therefore z = i \tan \theta &\Rightarrow z = i \tan\left(\pm \frac{\pi}{8}\right), i \tan\left(\pm \frac{3\pi}{8}\right) \\
\therefore z &= \pm i \tan\left(\frac{\pi}{8}\right), \pm i \tan\left(\frac{3\pi}{8}\right) \quad [\because \tan(-x) = -\tan(x)]
\end{aligned}$$

(d) (i)



Join QS and produce KH to intersect with QS at X .
Join RP

- (ii) (α) $PKRH$ is a cyclic quadrilateral (opposite angles are supplementary)
 $\angle PKH = \angle PRH$ (angles in the same segment)
 $\angle PRH = \angle PQS$ (angles in the same segment)
 $\therefore \angle PKH = \angle PQS$
- (β) $\angle PHK + \angle PKH = 90^\circ$ ($\because \angle KPH = 90^\circ$)
 $\angle QHX = \angle PHK$ (vertically opposite angles)
 $\therefore \angle QHX + \angle PQS = 90^\circ$ ($\because \angle PKH = \angle PQS$)
 $\therefore \angle QXH = 90^\circ$ (angle sum of Δ)
 $\therefore KH$ (produced) $\perp QS$

(e) If α is a real root of the equation $x^3 + ux + v = 0$ then $\alpha^3 + u\alpha + v = 0$

$$\text{Now } x^3 + ux + v = (x - \alpha)(x^2 + Ax + B)$$

$$\begin{array}{r} x^2 + \alpha x + (u + \alpha^2) \\ (x - \alpha) \overline{) x^3 + 0x^2 + ux + v} \\ \underline{x^2 - \alpha x^2} \\ (x - \alpha) \overline{) 0 + \alpha x^2 + ux} \\ \underline{\alpha x^2 - \alpha^2 x} \\ (x - \alpha) \overline{) 0 + (u + \alpha^2)x + v} \\ \underline{(u + \alpha^2)x - (u + \alpha^2)\alpha} \\ 0 \end{array}$$

$$\therefore v + (u + \alpha^2)\alpha = 0$$

$$\therefore x^3 + ux + v = (x - \alpha)[x^2 + \alpha x + (u + \alpha^2)]$$

With $x^2 + \alpha x + (u + \alpha^2) = 0$ to have real roots then

$$\Delta = \alpha^2 - 4(u + \alpha^2) = -3\alpha^2 - 4u \geq 0$$

$$\therefore 3\alpha^2 + 4u \leq 0$$